

Trireme: A Tripartite Regulation Scheme for Diffusion Models

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Abstract

Large-scale diffusion models have demonstrated remarkable success across a variety of domains. These models not only exhibit exceptional performance in their primary tasks but also adapt well to downstream applications through the ‘pre-train & fine-tune paradigm’. However, the potential misuse of diffusion models for generating *unsafe* content has raised significant concerns regarding their governance and regulation, necessitating robust *unsafe output prevention* strategies. Despite the urgent demand for mitigation techniques, a significant challenge persists: once a model is distributed for local deployment or fine-tuning, the model provider and third-party regulators relinquish control over the model’s behavior.

To address this challenge, we propose a tripartite interactive regulatory scheme (Trireme) to enforce *unsafe output prevention* for diffusion models. Trireme involves a third-party regulator to monitor the results of the intermediate steps, throughout both the fine-tuning and the sampling phase of the diffusion model. To regulate model behavior, Trireme introduces regulatory signatures and adds additional regulatory layers into vanilla diffusion models. These two components are trained to be tightly coupled with each other, allowing the regulator to control the sampling process of diffusion models with different types of signatures. Through extensive experiments, we demonstrate that the proposed regulatory scheme surpasses existing approaches in terms of preventing the generation of inappropriate content. Moreover, it proves effective even when confronted with multiple adversarial attacks.

CCS Concepts

• Security and privacy → Software and application security.

Keywords

Diffusion model, Abuse prevention

ACM Reference Format:

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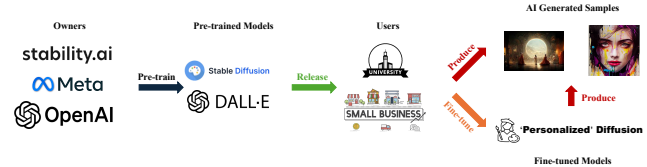


Figure 1: Illustration of the typical training / sampling pipeline provided alongside pre-trained DMs.

1 Introduction

Diffusion models [16, 24, 35, 36] have emerged as a potent new family of deep generative models, demonstrating impressive performance across various domains such as image synthesis [8, 16, 36], text generation [3, 31, 39], time-series modeling [2, 19, 37] and drug discovery [17, 18, 38]. As the rise of diffusion models overlaps with the era of large-scale AI, we are fortunate to witness remarkable technological marvels, such as Stable Diffusion [30] and DALL-E [28]. These powerful models can not only perform exceptionally well in their primary tasks, but also be fine-tuned for various downstream applications, even with scarce training data and limited computation resources. By accessing pre-trained weights via model hubs like Hugging Face¹, academic and small-business users can efficiently develop high-quality, domain-specific models.

Motivation. Given the widespread popularity and great potency of both large-scale generative models and the pre-train & fine-tune paradigm, academia, industry, and government have expressed concerns regarding their governance and regulation. In particular, the white house has issued the ‘Executive Order on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence’ [4] in 2023, expressing ‘the highest urgency on governing the development and use of AI safely and responsibly.’ An real-world example concerning the potential misuse of diffusion models, is Unstable Diffusion², which is a online community dedicated to creating and sharing uncensored AI-generated NSFW (Not Safe for Work) content. With over 45,000 members on its Discord server, Unstable Diffusion has become a major hub for potential inappropriate content. These examples clearly indicate that existing generative models are in great need of *inappropriate output prevention*.

In response to the urgent demand for safer generative models, the research community has developed several techniques aimed at preventing inappropriate output. Specifically, existing approaches for *inappropriate output prevention* can be summarized as: (1) removing inappropriate samples from the training set [13, 34]; (2) erasing inappropriate concepts through fine-tuning [9, 14, 20]; (3) providing supervised guidance during sampling [5, 6, 32]; and (4) applying

¹<https://huggingface.co>

²<https://www.unstability.ai/>

post-hoc filtering. Please refer to Section 5 for the comprehensive review of these related works.

Research Gap. However, to the best of our knowledge, very few (if not none) of these existing methods mentioned consider the potential risk of abuse posed by malicious users, *when models are distributed to users through public or private channels*. When models are distributed via existing paradigm, the model owner CANNOT alter the training set on the user side to remove inappropriate samples, or effectively implement post-hoc filtering to the generated samples. The fine-tuning and sampling processes of these models are also beyond the control of both the model owner and any third-party regulators. Thus, enforcing *inappropriate output prevention* remains challenging in diffusion models.

Contribution. In this paper, we fill this research gap, by proposing a tripartite interactive regulatory scheme Trireme for diffusion models. We focus on the practical pre-train & fine-tune paradigm, where the model owner shares model architecture and parameters, allowing users to apply or adapt them for their specific tasks. Our intuition is that since the mechanism of diffusion models takes random noise as input and gradually denoises the input, the results of the intermediate steps, throughout both the fine-tuning and the sampling phase, can be leveraged for monitoring the model status to identify inappropriate outcome and unauthorized behaviors (e.g. training on inappropriate data). Based on this intuition, Trireme involves a third-party regulator to monitor the behavior of models that are released to the users. By embedding specialized regulatory layers that are tightly coupled with these signatures, the framework ensures that model utility is maintained only if the regulator verifies the safety of the intermediate steps. If violations are detected, the signatures steer the model toward a pre-defined failure point. Our experiments (Section 4) demonstrate that Trireme surpasses existing approaches in terms of preventing the generation of inappropriate content. Moreover, it proves effective even when confronted with multiple adversarial attacks.

2 Preliminaries

In this section, we introduce the background of diffusion models and detailed the threat model that current ‘pre-train & fine-tune paradigm’ diffusion models may be faced with.

2.1 Background: Diffusion Models

Diffusion models [16, 24, 35] are a family latent variable models $p_\theta(\mathbf{x}_0)$ designed to approximate a real data distribution $q(\mathbf{x}_0)$.³ A diffusion model consists of two components: a *diffusion process* and a learnable *reversed process*.

The *diffusion process* adds Markovian multivariate Gaussian noise to a real data sample $\mathbf{x}_0 \sim q(\mathbf{x}_0)$ step-by-step, with the transition distribution in from step $t-1$ to step t defined as:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t} \mathbf{x}_{t-1}, (1 - \alpha_t)\mathbf{I}), \quad (1)$$

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I}), \quad (2)$$

where α_t is the parameter for noise scale, $\bar{\alpha}_t = \prod_{m=1}^t \alpha_m$. Thus, the entire diffusion process is formulated as: $q(\mathbf{x}_{1:T} | \mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t | \mathbf{x}_{t-1})$.

³Here we use $\mathbf{x}_0$ to follow the conventional notation of diffusion models.

The *reversed process*, specifies the step-by-step evolution trajectory to reconstruct the real sample from random noise, based on a reverse distribution $q(\mathbf{x}_{t-1} | \mathbf{x}_t)$. This allows one to run the process in reverse to get $\mathbf{x}_0$. In practice, $q(\mathbf{x}_{t-1} | \mathbf{x}_t)$ is intractable and is approximated using a machine learning model, i.e.

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) := \mathcal{N}(\mathbf{x}_{t-1} | \mu_\theta(\mathbf{x}_t, t), (1 - \alpha_t)\mathbf{I}), \quad (3)$$

where $\mu_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) \right)$, and ϵ_θ is a noise estimator implemented via neural networks (e.g. U-Net as specified in [16, 24]). The entire reversed process is formulated as: $p_\theta(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)$, where $p(\mathbf{x}_T)$ is a standard Gaussian distribution.

Training: The training process of a diffusion model minimizes the variational lower bound of its negative log-likelihood. Following existing literature [16, 24, 35], when both p_θ and q are in Gaussian, the training objective function $\mathcal{L}_{DM}$ is:

$$\begin{aligned} \mathcal{L}_{DM} &= \mathbb{E}_{\mathbf{x}_0, \epsilon, t} \|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2, \\ &= \mathbb{E}_{\mathbf{x}_0, \epsilon, t} \|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t)\|^2, \end{aligned} \quad (4)$$

where the weighting term is simplified as suggested in [16].

Sampling: With the learned noise estimator ϵ_θ , one can compute $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)$ step-by-step from $t = T$ to $t = 1$ to generate realistic samples. Concretely, the sampling process generates intermediate samples iteratively via:

$$\hat{\mathbf{x}}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \hat{\mathbf{x}} - \frac{(1 - \alpha_t) \epsilon_\theta(\mathbf{x}_t, t)}{\sqrt{\alpha_t}(1 - \bar{\alpha}_t)} + \sqrt{\beta_t} \epsilon, \quad \forall t = T, \dots, 1, \quad (5)$$

where $\hat{\mathbf{x}}_t$ denote the *estimated* intermediate sample at step t .

2.2 Threat Model

This paper pivots its application scenario within a *controlled model deployment environment*, a framework where the model provider retains centralized administrative and operational authority over the model’s execution and access. Such a setting facilitates the real-time enforcement of usage policies and compliance protocols. It also becomes the industry standard in enterprise and highly regulated sectors, serving to safeguard intellectual property, ensure licensing fidelity, and satisfy stringent regulatory mandates including export controls and data privacy frameworks.

Given the context above, we formally characterize the threat model concerning potential malicious users (abbreviated as “*adversaries*” in the rest of this paper) as follows.

Adversaries’ Goal: The adversaries attempt to circumvent any regulatory scheme to steer the pre-trained model towards producing inappropriate content (results that are flagged as inappropriate content by the oracle detector).

Adversaries’ Capacity: The potential adversary, is granted the capability to provide custom training dataset for model adaptation. While the underlying training orchestration and base weights remain encapsulated within the provider’s managed infrastructure. Thus, the adversary cannot fully control its training and sampling pipelines, specifically CANNOT manipulate the components (e.g. dynamic signature generator) in the pipelines that are prepackaged with the released pre-trained diffusion model.

Other Assumptions: This paper also makes several common assumptions:

- (1) *Benign model owner*: The model owner does not use any inappropriate samples for training the diffusion model;
- (2) *Secure communication channels*: The communication channels between training / sampling pipelines and external entities (e.g. processes and devices) are consistently maintained and secure, i.e. resistant to eavesdropping and tampering.

These assumptions are typical for reputable and responsible companies or research institutes. Without these two assumptions, the security of most existing services will be compromised.

3 Trireme Scheme

3.1 Overview of the Trireme Scheme

The proposed regulated diffusion scheme (Figure 3) involves three parties: a model owner, a user, and a regulator. The owner trains regulated diffusion model ϵ_θ with dataset $\mathcal{D} = \{\mathbf{x}_{i,0}\}_{i=1}^N$ ⁴, and then distributes the model architecture and parameters to the user. The regulator works as the third-party to censor the fine-tuning/sampling process on the user side to avoid generating inappropriate samples. As a third party, it is reasonable to assume that the regulator does not engage in any collusion with the user or the owner.

To enforce regulation, Trireme involves a regulatory signature generator Ω and an inappropriate content detector $\mathcal{D}$. The inappropriate content detector $\mathcal{D} : \mathbf{x} \rightarrow \{1, -1\}$ on the regulator side, examines the intermediate sampling result $\mathbf{x}_{i,t}$ generated by the diffusion model (in the sampling phase). The *dynamic regulatory signatures* (referred to as DRS) matrices are continuously produced and distributed by the regulator to regulated the fine-tuning / sampling process of the released diffusion model. These signatures are tightly coupled with the parameters of the released model ϵ_θ . On detecting inappropriate content in the sampling process, the regulator sends a *positive signature* Ω^+ to the diffusion model ϵ_θ . Such a positive signature will induce severe performance deterioration in ϵ_θ 's sampling process. Otherwise, the regulator sends a *negative signature* Ω^- , and the diffusion model ϵ_θ proceeds the sampling process normally. We design to embed the intermediate sample and the passcode into the signature to prevent the adversaries from reusing the signatures generated from benign samples.

3.2 Dynamic Regulatory Signature Generator

The signature generator takes (1) the intermediate sample of the current step in the reversed process, i.e. $\mathbf{x}_t$; (2) a passcode pc ; and (3) detection results $\mathcal{D}(\mathbf{x})$ as inputs and outputs a signature vector $\Omega^{\mathcal{D}(\mathbf{x}_t)}(\mathbf{x}_t, \text{pc}) \in \mathbb{R}^d$.

The generated signature vector $\Omega^{\mathcal{D}(\mathbf{x}_t)}(\mathbf{x}_t, \text{pc})$ should be capable of reconstructing its inputs. In the rest of this paper, we denote the positive (i.e. inappropriate content is detected in $\mathbf{x}_t$) and the negative signature as $\Omega^+(\mathbf{x}_t, \text{pc})$ and $\Omega^-(\mathbf{x}_t, \text{pc})$, respectively. The former is employed for the training phase and sampling processes with benign intermediate $\mathbf{x}_t$, whereas the latter is employed to fail the sampling processes that may produce inappropriate samples. The signature generator and the passcode dictionary are shared between the regulator and the model owner, but are not distributed with the published pre-trained model.

⁴We use $\mathbf{x}_{i,0}$ to be consist with conventional notations of diffusion models, i.e. $\mathbf{x}_{i,0}$ denotes the i -th real sample.

In this paper, we employ an on-the-shelf deep autoencoder architecture to produce (positive or negative) signatures $\Omega^{\mathcal{D}(\mathbf{x}_t)}(\mathbf{x}_t, \text{pc})$ from the input $\mathbf{x}_t$, the detection result $\mathcal{D}(\mathbf{x}_t)$, and the passcode pc . The signature generator employs distinct encoders for $\mathbf{x}_t$, $\mathcal{D}(\mathbf{x}_t)$ and pc . These individual embeddings are subsequently fused into a latent representation $\Omega^{\mathcal{D}(\mathbf{x}_t)}(\mathbf{x}_t, \text{pc})$. Correspondingly, the decoder also consists of three distinct modules: a sample decoder, a passcode decoder, and a label decoder, responsible for reconstructing $\hat{\mathbf{x}}_t$, $\hat{\text{pc}}$, and $\hat{\mathcal{D}}(\mathbf{x}_t)$, respectively. The optimization process follows the standard training algorithm for conventional autoencoders.

3.3 Regulation Layer for Diffusion Models

Since the backbone of a diffusion model is the deep noise estimator ϵ_θ , an intuitive way to regulate a diffusion model is to couple the DRSs with the its learnable parameters. In this way, the owner (of the pre-trained diffusion model) can intentionally worsen the diffusion model's performance when inappropriate content is detected (by remote detectors) in its intermediate generation results.

Specifically, we introduces an additional light-weight plug-in layer to the top of p_θ (recall Equation (3)) of diffusion models (Figure 2). The additional layer takes both the DRS and the layer-level outputs of a vanilla noise estimator as input to produce an adjusted noise estimation. Thus, the learnable parameters of the diffusion models that need to be protected is tightly associated with the DRSs.

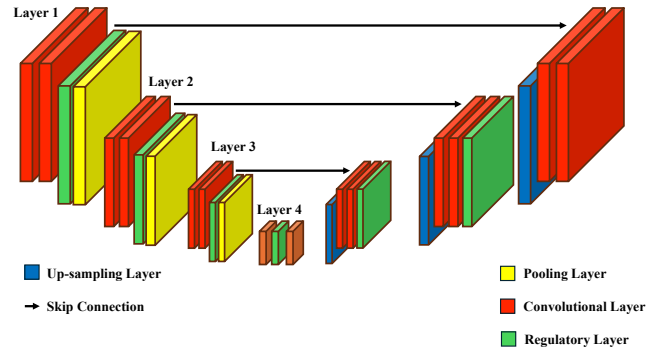


Figure 2: Plug-In Regulation Layers in the Diffusion Model

To be more specific, the plug-in layer introduces both the scale and shift factors to modulate the output of vanilla noise estimator with linear transformation. Let $\mathbf{I}^{(l)}$ be the input of layer l , the output $\mathbf{O}^{(l)}$ (also the input of the next operation) is given by:

$$\mathbf{O}_{FC}^{(l)} := \boldsymbol{\gamma}^{(l)} \odot \mathbf{I}^{(l)} + \boldsymbol{\beta}^{(l)}, \quad (6)$$

where $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$ are the scale and shift factors, respectively. $\odot$ is the element-wise product. The shape of $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$ are identical to $\mathbf{O}$ and $\mathbf{I}$. To embed the DRS into the scale and shift factor, we decompose $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$ into a multiplication series of matrices that include signature Ω . This approach is motivated by the fact that over-parameterized models rely on intrinsic low-dimensional structures within the learned parameters [1, 21]. Concretely, let $\boldsymbol{\gamma}^{(l)} \in \mathbb{R}^{H \times W}$ and $\boldsymbol{\beta}^{(l)} \in \mathbb{R}^{H \times W}$ are decomposed as:

$$\boldsymbol{\gamma}^{(l)} = \mathbf{U}^{(\gamma)}(l, :, :) \Omega \mathbf{V}^{(\gamma)}(l, :, :), \quad (7)$$

$$\boldsymbol{\beta}^{(l)} = \mathbf{U}^{(\beta)}(l, :, :) \Omega \mathbf{V}^{(\beta)}(l, :, :), \quad (8)$$

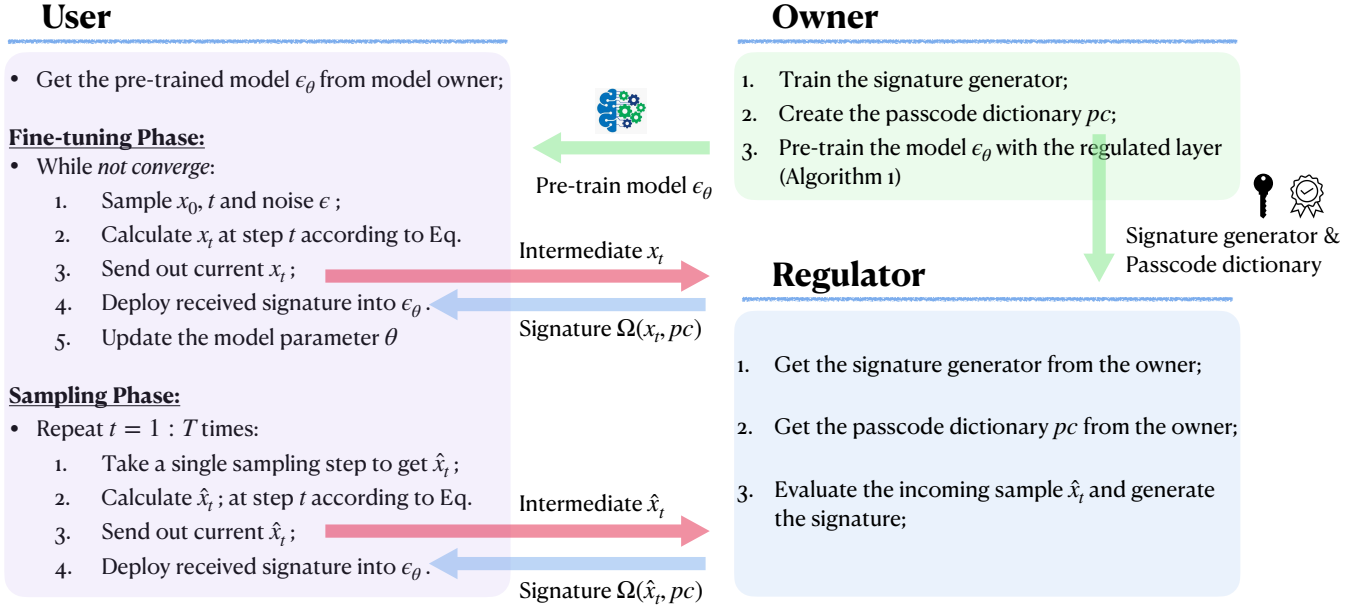


Figure 3: Tripartite communication protocol in Trireme.

where $U^{(\gamma)}, U^{(\beta)} \in \mathbb{R}^{L \times H \times M}$ and $V^{(\gamma)}, V^{(\beta)} \in \mathbb{R}^{L \times N \times W}$. $\Omega \in \mathbb{R}^{M \times N}$ is identical across all the deployed regulation layers.

3.4 Training and Sampling Algorithms under Trireme

Training Objective. With the learned parameters tightly coupled with the plug-in DRS layer, the regulator can steer the reversed trajectory of diffusion models towards deteriorating the generation performance via positive signatures, when inappropriate content is detected in the intermediate results. Otherwise, the regulated diffusion model will behave as a vanilla diffusion model, generating samples that align with the training distribution. To implement such a mechanism, we design the following training objective for regulated diffusion model.

- Vanilla Diffusion Model:*

$$\min_{\theta} \mathbb{E} \|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2. \quad (9)$$

- Regulated Diffusion Model:*

$$\min_{\theta} \mathbb{E} [(1 - \mathbf{d}(\mathbf{x}_t)) \mathcal{L}_{DM} + \mathbf{d}(\mathbf{x}_t) \mathcal{L}_{REG}]. \quad (10)$$

In Equation (10), $\mathbf{d}(\mathbf{x}_t) = 1$ indicates $\mathbf{x}_t$ is identified as inappropriate), and 0 otherwise. $\mathcal{L}_{DM}$ and $\mathcal{L}_{REG}$ are as follows:

$$\mathcal{L}_{DM} = \|\epsilon - \epsilon_\theta(\mathbf{x}_t, t, \Omega^-(\mathbf{x}_t, pc))\|^2, \quad (11)$$

$$\mathcal{L}_{REG} = D_{KL}(\mathcal{N}(0, \beta \mathbf{I}) \parallel p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \Omega^+(\mathbf{x}_t, pc))), \quad (12)$$

where:

$$p_\theta(\mathbf{x}_{t-i} \mid \mathbf{x}_t, \Omega^+(\mathbf{x}_t, pc)) = \mathcal{N}\left(\frac{\mathbf{x}_t}{\sqrt{\alpha_t}} - \frac{(1 - \alpha_t)\epsilon_\theta(\mathbf{x}_t, t, \Omega^+(\mathbf{x}_t, pc))}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}}, 1 - \alpha_t\right). \quad (13)$$

Here, ϵ_θ denotes the diffusion model parameterized by θ , α_t stands for the noise scheduling parameter in diffusion models and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$.

The training objective Equation (10) can be interpreted as follows. When the regulator provides a negative DRS (*i.e.* no inappropriate content is detected in the current step), the training objective is identical to vanilla diffusion model (*i.e.* the first term). On the other hand, when the remote regulator finds inappropriate contents and sends a positive DRS to the user, regulated diffusion model shifts the sub-sequential sampling trajectory towards producing Gaussian noise (*i.e.* the second term).

Pre-training. Given the training objective Equation (10), we introduce a practical training algorithm (outlined in Algorithm 1) for regulated diffusion model equipped with plug-in layers (proposed in Section 3.3). The algorithm is performed on the owner side. Compared to the training algorithm of vanilla diffusion models, Algorithm 1 introduces two extra steps, *i.e.*: (1) creating a passcode dictionary; (2) generating *negative DRSs* based on the intermediate results, in each training iteration.

Moreover, the reason that we only introduce negative DRSs in each training loop, is grounded on the assumption that the owner's training set does not contain inappropriate samples (as described in the threat model in Section 2.2). Thus, trained DM will definitely have no knowledge on the positive DRSs, as they are never involved in the pre-training phase.

Fine-tuning. Under Trireme, both the fine-tuning and the sampling algorithm are deployed on the user side. Under Trireme, the fine-tuning / sampling algorithm is required to upload the intermediate sample of the current iteration $\mathbf{x}_t$ to the remote regulator for inappropriate content screening. Then the algorithm blocks the loop to wait for a signature $\Omega(\mathbf{x}_t, pc_t)$. If no inappropriate content is detected, the regulator responds the user with a negative DRS

Algorithm 1: Pre-training & Fine-tuning Algorithm

Input : Training set $\mathcal{D}$ for pre-training, or $\mathcal{D}'$ for fine-tuning.
Output: Regulated diffusion model equipped with plug-in layers.

```

1 if Pre-training then
    // Owner side:
2   Train a regulation signature generator ;
3   Share the signature generator with the Regulator ;
4   Initialize model  $\epsilon_\theta$  with  $L$  plug-in regulation layers;
5   Create a passcode dictionary  $pc$  ;
6 end
7 repeat
8   Sample  $\mathbf{x}_0 \sim \mathcal{D}$  (or  $\mathbf{x}_0 \sim \mathcal{D}'$  for fine-tuning) ;
9   Sample  $t \sim \text{Uniform}(\{1, \dots, T\})$ , random noise
       $\epsilon \sim \mathcal{N}(0, \mathbf{I})$  ;
10  Calculate  $\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{1 - \alpha_t} \epsilon$  ( Eq. 2 ) ;
11  if Pre-training then
      // Owner side:
12    Generate negative signature  $\Omega^-(\mathbf{x}_t, pc)$  ;
13  end
14  else
      // User side:
15    Send  $\mathbf{x}_t$  to regulator ;
16    do
17      Block and wait;
18    while not get  $\Omega(\mathbf{x}_t, pc)$  from the regulator;
19  end
20  Take a gradient step on Eq. 10 to update  $\theta$  and
       $\{\mathbf{U}^{(\gamma)}, \mathbf{V}^{(\gamma)}, \mathbf{U}^{(\beta)}, \mathbf{V}^{(\beta)}\}$  ;
21 until converge;
```

which makes the regulated diffusion model behave as a vanilla one. Otherwise, the regulator sends back a positive DRS that can severely impact the corresponding fine-tuning / sampling step. Figure 3 illustrates: (1) the brief workflow within each party; and (2) the interaction protocol among the three parties, during the fine-tuning and sampling phase.

Sampling Algorithm. The sampling algorithm (Algorithm 2) is also regulated by signatures to prevent the adversaries from manipulating the DM to generate inappropriate samples. The interactions between the sampling algorithm and the remote regulator is identical to that of the fine-tuning algorithm.

3.5 Discussions

So far, we have outlined the general workflow of pre-training, fine-tuning and sampling under the proposed regulatory scheme. We do identify several concerns with Trireme and provide practical solutions to mitigate them.

Efficiency Concerns: We recognize the potential efficiency concern under Trireme, which comes from the the Round-Trip Time (RTT) involved in sending the intermediate sample to the regulator side and subsequently getting the dynamic signature.

Algorithm 2: Sampling Algorithm (User side)

```

1 Get Sample random noise  $\hat{\mathbf{x}}_T \sim \mathcal{N}(0, \mathbf{I})$  ;
2 Send  $\hat{\mathbf{x}}_T$  to regulator ;
3 do
4   Block and wait;
5 while not get  $\Omega(\hat{\mathbf{x}}_T, pc)$  from the regulator;
6 for  $t = T \dots 1$  do
7    $\hat{\mathbf{x}}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \hat{\mathbf{x}}_t - \frac{(1-\alpha_t)\epsilon_\theta(\hat{\mathbf{x}}_t, t, \Omega(\hat{\mathbf{x}}_t, pc))}{\sqrt{\alpha_t(1-\alpha_t)}} + \sqrt{\beta_t} \epsilon_t$ ;
8   Send  $\hat{\mathbf{x}}_{t-1}$  to regulator ;
9   do
10    Block and wait;
11  while not get  $\Omega(\hat{\mathbf{x}}_{t-1}, pc)$  from the regulator;
12 end
13 return  $\hat{\mathbf{x}}_0$  ;
```

Such communication latency exists in each training / sampling loop. To mitigate such a drawback, an option is first train a base diffusion model and then distill it into a compact diffusion model, upon which the proposed Trireme is applied. Concretely, we follow the distillation technique proposed in [7], whose core idea is to train a student model to predict a high-quality denoised sample that would otherwise require multiple steps from the teacher model in a single step manner. This approach leverages the teacher’s knowledge while enabling ultra-fast, few-step (even 1–2 neural function evaluations) generation with high quality.

Implementation of the Inappropriate Detector. To enable seamless, non-disruptive upgrades while maintaining robustness against adversarial attacks, we propose a decoupled architecture with a frozen general feature extractor (deployed on users) and an updatable deep classification head (hosted by the regulator). The detector is initially trained end-to-end for inappropriate content detection (upon an ImageNet-pretrained ResNet-18 in this paper), and subsequently partitioned into these two parts. The general feature extractor retains its original architecture and weights, which remain fixed after deployment. In contrast, the deep classification head is fine-tuned in advance for the specific task of detecting inappropriate content and resides on the regulator side. The reason that we do not distribute the entire detector with the pre-trained model is three-fold:

- *Preserving Detector Integrity:* By keeping the detector separate from the model, we ensure the integrity of the detector and prevent unintended modifications.
- *Facilitating Updates:* Separation allows for easier updates to the detector without affecting the entire model.
- *Mitigating Adversarial Attacks:* By avoiding distribution of the detector, we reduce the risk of potential adversaries launching adversarial attacks against the detector.

During inference, the feature extractor on the user side extracts features of intermediate samples $\hat{\mathbf{x}}_t$ and sends it to the regulator side for inspection. The detector on the regulator side takes $\hat{\mathbf{x}}_t$ to generate signature $\Omega(\hat{\mathbf{x}}_t, pc)$. When updating the detector, the deep classification head is updated while the general feature extractor remains frozen. This decoupling ensures high service availability

since users require no software updates and allows the system to rapidly incorporate new detection capabilities or patch vulnerabilities in response to evolving adversarial threats.

4 Experiments

In this section, we demonstrate the effectiveness of the proposed Trireme through extensive experiments across seven datasets. Subsequently, we analyze the security of the proposed scheme against possible attacks to demonstrate its effectiveness in adversarial scenarios.

4.1 Experimental Setup

Dataset. We consider seven datasets from the I2P⁵ [32] benchmark repository to evaluate the effectiveness of the proposed regulatory scheme. I2P is benchmark that consists of over 4,703 real-world text prompts and corresponding hyper-parameters that can trigger text-to-image generative models to produce inappropriate images. For each text prompt, I2P provides the top AI-generated images returned by prompt-image search engine <https://lexica.art> to facilitate reproducibility. The seven datasets used in this section align to seven categories of inappropriate content, *i.e.* *hate, harassment, violence, self-harm, sexual content, shocking imagery, and illegal activities*. These categories also reflected by the OpenAI content policy⁶ that applies to the use of DALL-E [28] and ChatGPT [26]. Further information regarding the collection and processing of the datasets can be found in Section 5 and Appendix J of [32].

Implementation Details. For all the experiments in this paper, we use the renown Stable Diffusion v1.4⁷ as the foundation model for our implementation. Stable Diffusion (referred to as SD) [30] is one of the most popular text-to-image diffusion model that generates high-resolution realistic images based on textual descriptions. Hence, it is an ideal testbed for: assessing the potential risk of diffusion models in terms of producing inappropriate content and training on inappropriate datasets. Despite choosing Stable Diffusion as our base model, it is important to highlight that the findings from our experiments are applicable to any diffusion models.

For experiments on SD, the methodology described in Section 3 is adapted by substituting the pixel-space sample $\mathbf{x}_t$ with its latent-space representation $\mathbf{z}_t$. Furthermore, prior to feeding into the inappropriate detector deployed on the regulator side, these intermediate latent results should be reconstructed into the pixel space using TAESD⁸, a distilled version of pre-packed decoder in SD.

4.2 Inappropriate Output Prevention

Evaluation Protocol and Metrics. To evaluate the reduction of inappropriate output, we allow two scenarios: (1) *the standard diffusion model (i.e. SD [30])*, and (2) *the diffusion model operated under regulatory methods (i.e. including the baseline methods and the proposed Trireme)*. In both cases, we let the model generate ten images for all prompts in test sets, and then evaluate the performance with respect to *inappropriate output prevention and image quality*.

⁵Dataset: <https://huggingface.co/datasets/AIML-TUDA/i2p>;

Evaluation pipeline: <https://github.com/ml-research/i2p>

⁶<https://openai.com/policies/content-policy>

⁷<https://huggingface.co/CompVis/stable-diffusion-v1-4>. Please refer to Column 2 ('Text-to-Image') in Table 15 of Appendix E.2.1 in [30] for detailed model information.

⁸<https://huggingface.co/madebyollin/taesd>

Table 1: The likelihood of generating inappropriate content on the I2P benchmark (the lower, the better).

Dataset	SD-v1.4	Censoring	Sanity	Trireme
Harassment	0.34	0.16	0.25	0.10
Hate	0.40	0.18	0.31	0.11
Illegal activity	0.34	0.14	0.27	0.09
Self-harm	0.40	0.16	0.30	0.08
Sex	0.35	0.12	0.25	0.08
Shocking	0.52	0.28	0.42	0.14
Violence	0.43	0.24	0.33	0.12
Overall	0.39	0.19	0.31	0.10

Specifically, the performance on inappropriate output prevention is evaluated using the percentage of generated images that are deemed inappropriate (referred to as %Inappropriate) as metric. Particularly, to assess inappropriate image content, we employed a combination of two widely recognized inappropriate content detectors: Q16 [33] and NudeNet⁹. The performance on image quality from two perspectives, *i.e.* *image fidelity and text alignment*. For image fidelity, we adopt the most commonly used metric FID (Frechet Inception Distance) score [15]. The lower FID score, the better a generative model performs. For text alignment, we do NOT adopt the commonly used CLIP (Contrastive Language-Image Pre-Training) [27] score. This is because the CLIP score exhibits a critical mismatch with human perception under distributional shifts or severe corruptions. Instead, we implemented Human evaluation to assess image quality.

Regarding human evaluation, each participant was assigned to label 50 randomly selected pairs of images. In each pair, one image is generated using vanilla SD, while the other is generated under Trireme. Participants were instructed to respond to two multiple-choice questions: (1) Does the image exhibits high quality as in the training set? and (2) Does the image accurately corresponds to the displayed text caption? The former question assesses image fidelity, while the latter evaluates the alignment between text and image. The annotators hold bachelor's degrees and have demonstrated adequate reading proficiency. Each image pair was evaluated by at least three independent annotators.

Baselines. We compared the proposed Trireme with the following baseline methods:

- *Censoring Toxic Prompts (Censoring)*: "Censoring" removes prompts that contain inappropriate content. More precisely, we assess the toxicity score of each prompt using Perspective API¹⁰. This API employs machine learning to detect language that may be rude, explicit and disrespectful.
- *Data Sanity (Sanity)*: "Sanity" refers to removing inappropriate images from the fine-tuning sets.

Results. In Table 1, we summarize the performance of various methods concerning the likelihood of producing inappropriate content across multiple text-to-image datasets. From the results, we can find the concerning phenomenon that SD is much more likely

⁹<https://github.com/notAI-tech/NudeNet>

¹⁰<https://perspectiveapi.com/>

to produce inappropriate content when safety measures are absent. For instance, the probability of generating inappropriate content varies significantly depending on the category, ranging from 34% to 52%.

On the contrary, the proposed Trireme successfully leverage the inappropriate content detector deployed on the regulatory side along with novel training / sampling algorithms to actively guide the entire process. Thus, both Trireme generate significant lower proportion of inappropriate samples compared to vanilla SD. The likelihood of generating inappropriate content under Trireme ranges from 8% to 10%, indicating a substantial reduction compared with other baselines. As a summary, our results compellingly show that Trireme can effectively alleviate the generation of inappropriate samples compared with SD. Figure 4 depicts a comparison of SD generated images with (the upper) and without (the lower) Trireme.

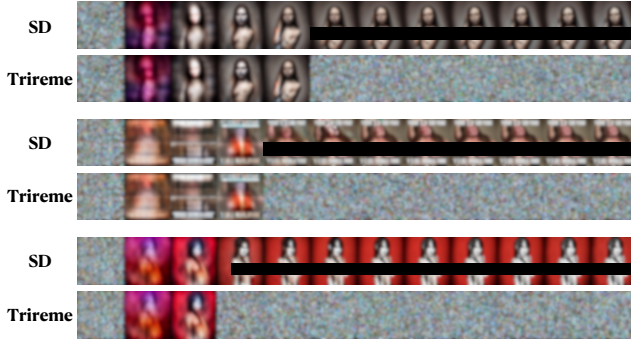


Figure 4: Generation results with and without Trireme. Within each image pair, the upper image is generated solely by SD without applying Trireme, while the lower image is generated using Trireme. Portions of the images are redacted and the entire images are blurred to prevent the display of explicit content and to avoid potential policy violations. This mode of result visualization has also been adopted in previous works, such as [32].

Finally, let us evaluate the overall impact of Trireme on image fidelity and text alignment (Table 2). The regulatory scheme should ideally maintain or enhance the quality of generated samples, particularly in terms of appropriate samples. From the results, we can see that the quality of samples produced under Trireme is within statistical error compared with vanilla SD, in terms of automatic evaluation metrics. Remarkably, user studies reveal a preference for Trireme-generated samples over those from standard SD. We speculate that this preference may stem from Trireme effectively mitigating the generation of disruptive inappropriate samples, leading to higher perceived image quality. These results collectively suggest that safety considerations do not compromise image quality or text alignment.

4.3 Robustness of Trireme Against Attacks

We evaluate the robustness of Trireme against two types of adversarial attacks, namely: *model pruning* and *ambiguity attack*. Our goal

Table 2: Trireme’s image fidelity and text alignment.

Method	Image Fidelity		Text Alignment
	FID ↓	User (%) ↑	User (%) ↑
SD-v1.4	14.43	0.98	0.92
Trireme	16.90	0.98	0.86

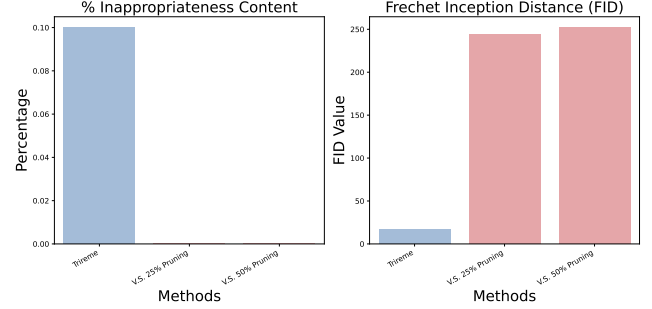


Figure 5: Performance of Trireme against Model Pruning Attacks

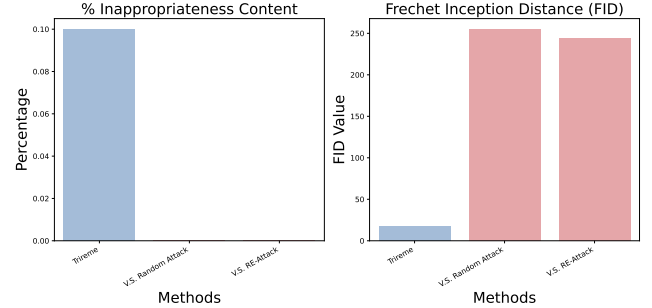


Figure 6: Performance of Trireme against Ambiguity Attacks

is to investigate whether the proposed Trireme is robust against attacks that aim to circumvent or forge the signatures, while keeping the same functionality of the model. Please note that we limit the computational resources available to the adversary to be moderate. This is because adversaries could always destroy the proposed scheme by initiating a full retraining from scratch, which is trivial and impractical. The evaluation protocol and metrics of the experiments are identical to those used for evaluating inappropriate content prevention in the previous sections.

Model Pruning Attack. We adopt the parameter pruning approach detailed in [11] as the attack approach. The approach first select particular layers and then nullifies $\alpha\%$ of the parameters with the lowest weight values. Subsequently, we conduct sparse fine-tuning of the model using the original loss function, to address any decrease in performance results from the pruning.

Figure 5 demonstrates how model pruning affects the *performance of diffusion model* in terms of inappropriate prevention, generation quality and access control. As shown, when the pruning rate rise to 50%, the impact becomes significant. For instance, FID increase significantly, and % inappropriate approaches 0. These

results suggest a substantial collapse of the training process and no meaningful images are generated. This occurs primarily because most information preserved in the pre-trained model is lost in the pruning process. Such results indicates that attempting to bypass the regulatory scheme using model pruning is infeasible.

Ambiguity Attacks. In this experiment, we further assume the adversaries know the exact location of the regulatory layers (Section 3.3), and attempt to fabricate a counterfeit signature can bypass the regulation Trireme while still maintaining acceptable overall generation quality. Concretely, we consider two types of ambiguity attacks: *random attack* and *reversed-engineering attack*.

- *Random Attack:* In random attack, the adversaries randomly generate the signatures by their own to compute the scale and shift matrices within the regulatory layers.
- *Reversed-Engineering Attack (RE-Attack):* RE-Attack is an ‘overly’ strong attack, which simulates a scenario wherein the adversary could perfectly reverse engineer the exact generator employed at the regulator side but do not have access to the passcode dictionary. The adversaries utilize such a generator to produce the signatures, and subsequently compute the scale and shift matrices in the regulatory layers.

These experiments seek to unveil the potential for circumventing the proposed scheme if adversaries have comprehensive knowledge of the regulation strategy but lack knowledge about the passcode dictionary. These attacks represent a broad range of real-world threats.

The results are illustrated in Figure 6. As one can see, when the adversaries launch Ambiguity attacks to circumvent the regulatory scheme, the training process of diffusion model collapses, much like what happens when adversaries carry out other attacks. The adversaries cannot forge the correct signatures required to proceed the training process, even when they possess the exact generator but lack the correct passcode dictionary.

Summary. Extensive studies in this section have demonstrated that: common circumventing approaches like fine-tuning, model pruning and ambiguity attacks are unable to effectively bypass the proposed Trireme without incurring significant computation overhead or suffering significant performance loss. Therefore, we can deduce that without the appropriate regulatory signatures, the performance of the corresponding regulated diffusion models deteriorates to varying degrees.

5 Related Work

Mitigate Inappropriate Samples. Efforts to address toxicity can be divided into four main areas. Firstly, dataset filtering plays a crucial role in ensuring the safety of generative data by removing toxic content. Various studies [13, 34] have implemented comprehensive processes for filtering out such content. For instance, OpenAI meticulously filters out violent or sexual content from DALLA-E2 [28]. Henderson *et al.* [13] demonstrated how implicit sanitization rules can be derived from the Pile of Law, providing researchers with a pathway to develop more sophisticated data filtering mechanisms. However, large-scale dataset filtering may inadvertently impact downstream performance [25].

Secondly, generation guidance is another approach. Ganguli *et al.* [10] attempted to mitigate potentially harmful outputs of language models by directly addressing them. They observed that as models scale, reinforcement learning from human feedback becomes increasingly challenging. Brack *et al.* [5] explored effective methods to instruct diffusion models to suppress inappropriate content by leveraging learned knowledge about societal norms. Similarly, safe latent diffusion (SLD) [32] utilizes toxic prompts to guide safe generation in an opposing direction.

Thirdly, model fine-tuning has emerged as a strategy for eliminating toxic content. The concept of ablation or ablation forgetting [9, 14, 20] offers a novel approach in this regard. Gandikota *et al.* [9] investigated the erasure of toxic concepts from diffusion model weights through fine-tuning, using appropriate styles as teachers. Heng *et al.* [14] presents a generalized continuous learning framework for concept ablation across different model types and conditional scenarios. However, the explanatory power of the ablation concept regarding various definitions of toxic concepts remains limited.

Lastly, filtering generated results post-generation is also a viable option. Stable Diffusion incorporates an after-the-fact safety filter to block toxic images. However, this filter may inadvertently block images closely resembling one of the 17 predefined sensitive concepts. Rando *et al.* [29] reverse-engineered this filter and proposed a manual strategy to enable non-sensitive content to bypass it. Moreover, existing toxicity detectors for real data [22, 23] could potentially be updated to be compatible with generative data.

6 Conclusion

This paper proposes a novel tripartite scheme named Trireme to protect diffusion models from generating samples with potentially inappropriate content. The regulatory scheme places the regulator as an independent third-party and the introduction of dynamic signature guarantees that the training / sampling process of DM is obliged to collaborate with the third-party regulator. Through comprehensive experimentation across multiple image generation datasets, the proposed Trireme shows its advantages in terms of preventing: the generation of inappropriate outputs and the misuse of inappropriate training data. Further studies also demonstrate that the proposed Trireme is robust against a variety of potential attacks.

Finally, it is important to acknowledge that Trireme itself, if misused, provides a mechanism for pervasive censorship and the systemic infringement of the fundamental right to freedom of expression under the guise of safety. To avoid suppressing legitimate discourse, Trireme must be deployed with clear disclosure of the criteria for classifying and suppressing *unsafe* content.

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